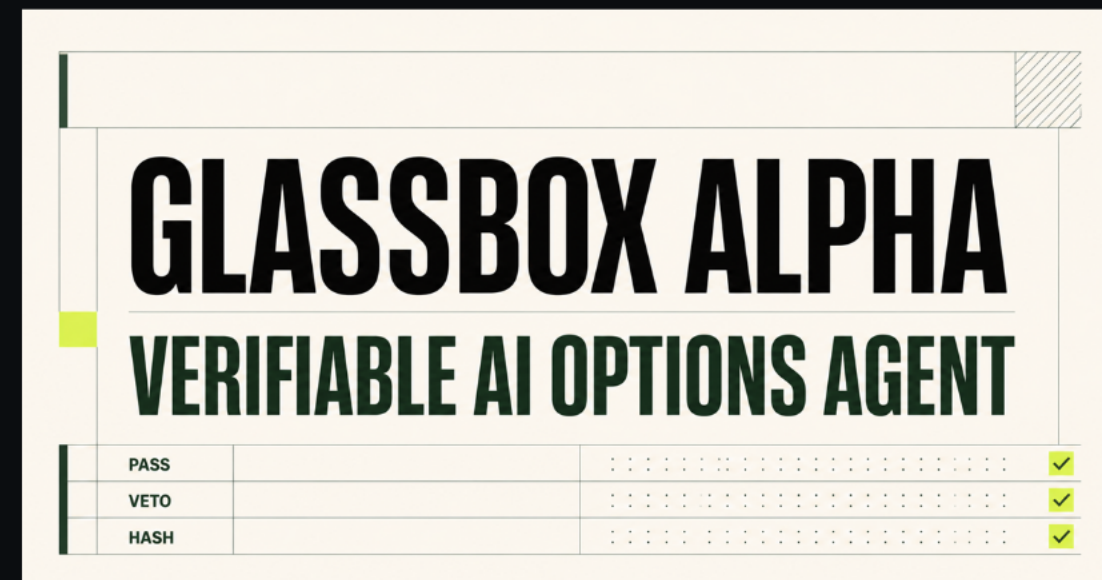


GlassBox Alpha

A verifiable AI options agent

**AI may veto.
It can never change the trade.**



PAPER ONLY

SPY / QQQ defined-risk options
32 deterministic checks
Hash-chained Trade Passports

The core risk is unrestricted model authority

Typical agent

The model can choose the ticker, size, price, and action in one step.

A persuasive output can become an execution decision before independent controls verify it.

GlassBox Alpha

Code creates one immutable candidate. AI receives veto-only authority.

A separate risk kernel retains the final decision and records every observed boundary.

Probabilistic judgment. Deterministic authority.

One candidate passes through seven controlled stages



The candidate is hashed before AI review and checked again immediately before execution.

Completed bars only • fixed economics • idempotent order identity

The AI critic can only reduce action

ALLOW

or

VETO

No trade editing. No execution tool.

Immutable	symbol / strikes / expiry / size / price
------------------	--

Evidence-bound	all cited evidence IDs must be provided
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Fail-closed	timeout / bad schema / mutation = VETO
--------------------	--

Rechecked	deterministic limits run after the model
------------------	--

Thirty-two checks run after the model

Each gate records its observed value, limit, outcome, and explanation.

32

deterministic gates

identity, economics, structure,
liquidity and state

1.00%

max options exposure

default portfolio boundary before a
new entry

0

live trading paths

paper mode is enforced in
configuration and execution

Any failure produces ABSTAIN. Unknown order state engages the kill switch.

Alpaca execution stays paper-only and atomic

READ

Account, clock,
completed bars,
contracts, chain
snapshots and positions

alpaca-py



SUBMIT

One atomic MLeg limit
order through the official
Alpaca CLI

paper endpoint only



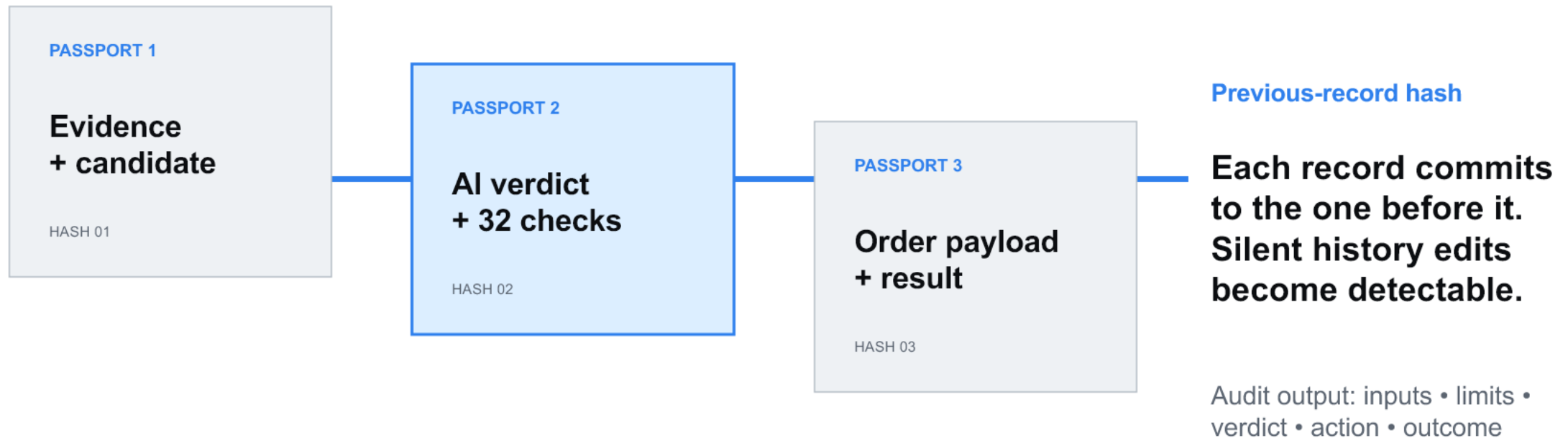
SUPERVISE

Reconcile exact
positions and close the
whole structure together

profit • loss • time • signal

Ambiguous broker response → persistent kill switch → manual reconciliation

Every decision becomes a tamper-evident Trade Passport



Judges can reproduce approval and abstention

CLEAN		APPROVED PREVIEW	24 / 24 automated tests VALID hash chain
AI VETO		ABSTAIN	
STALE QUOTE		ABSTAIN	
WIDE SPREAD		ABSTAIN	

Not a promise of returns — a verifiable boundary for financial AI.